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Swami Keshvanand Institute of Technology, Management & Gramothan. Ramnagaria, Jagatpura, Jaipur
Department of Computer Science & Engineering
Project (2026-27)

Functional Requirements Finalization

Project ID: SKIT/CSE/2023-2027/26 **Branch:** CSE **Section:** B

Project Title: Secure Cred - Blockchain Based Digital Certification System

SDG Mapping: SDG 4 (Quality Education)

Project Overview: Provide a brief description of the project (2-3 sentences)

SecureCred is a Blockchain Based Digital Certification System that enables authorized institution to issue tamper proof academic and vocational certificates. Each certificate is assigned a unique ID and a SHA-256 hash, stored via decentralized IPFS / Pinata Storage, and anchored on the Polygon blockchain though a Solidity smart contract for instant trustworthy verification.

Objectives: Main objectives of the project

1) Objective1 To design and implement Solidity smart contract for secure certificate issuance, on chain verification, and revocation on the Polygon blockchain.
2) Objective2 To develop backend REST APIs with PostgreSQL for certificate metadata storage, and to implement role based authentication and access control.
3) Objective3 To integrate decentralized IPFS / Pinata storage for certificate file and generate SHA-256 cryptographic hashes for temper Detection.
4) Objective4 To build institution, student and verifier dashboards enabling certificate issuance, viewing and QR / ID - based verification.

5) Objective5. To ensure system security and reliability through Smart Contract audits, API security hardening and end to end integration testing before deployment

FunctionalRequirements:

Detail the specific functional requirements of the project.

ID	Requirement Description	Priority (High/ Medium/Low)
FR-001	The system shall allow authorized institution to enter certificate and student details and issue digital certificates through the platform	High
FR-002	The system shall generate a unique certificate ID and SHA-256 hash, store the certificate on IPFS/Pinata and anchor its hash.	High
FR-003	The system shall allow external verifiers to verify a certificate using a certificate ID or QR code, checking the certificate information.	High
FR-004	The system shall allow authorized insitution to revoke an issued certificate, updating its status on the blockchain so that verifies are shown the correct revoke.	Medium.
FR-005	The System shall provide role-based dashboards for institution, studend and administration to manage certificate	Medium

Non- Functional Requirements:

Detail the specific functional requirements of the project.

ID	Requirement Description	Priority (High/ Medium/Low)
NFR-001	Only authenticated user acting within their assigned role shall be able to access role specified functionality	High
NFR-002	• The system shall detect certificate tampering by comparing its hash with the immutable blockchain record	High

NFR-003	The certificate verification process shall be return a valid/revoked result within 3-5 seconds under normal network condition	Medium
NFR-004	Issued Certificate file shall remain available for retrieval at least 99% of the time, independent of wheather the issuing institute own system are online	Medium
NFR-005	The system shall support deployment across environment and scale to handle increasing Certificate issuance and verification request.	Low

Assumptions and Constraints

1) Assumption1 All authorized institution using the platform will provide accurate certificate and student details at the time of issuance.
2) Assumption 2 The blockchain network (Polygon) and IPFS/ Pinata storage service will remain accessible and operational throughout development and deployment.
3) Constraint 1 The project anchors certificate hashes only on the Polygon blockchain, using its testnet during development and mainnet for final deployment.
4) Constraint 2 The system depends on third-party IPFS/Pinata services for decentralized file storage, so their availability directly affects certificate storage and retrieval

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